

Day 16 (25/11/2025)

Recursion functions -

A recursive function is a function that calls itself until a base condition is met.

base case: condition to stop recursion.

recursive case: functions calls itself with modified in

ex -

```
def factorial(n):
```

```
    if n==1:
```

```
        return 1
```

```
    else:
```

```
        return n * factorial(n-1)
```

factorial(5)

120

5 * 4

20 * 3

60 * 2

120

fibonacci series -

0
1
1
2
3
8

=>

0, 1, 1, 2, 3, 8

+ + + +

a=0
b=1

```
for i in range(1,6,1):  
    print(a,end=" ")  
    a,b=b,a+b
```

0 1 2 3

a=0

b=1

i=1

while

i<=5:

print(a,end=" ")

a,b=b,a+b

i=i+1

0 1 2 3

```
def fib(n):
```

```
    if n<=1:
```

```
        return n
```

```
    else:
```

```
        return fib(n-1) + fib(n-2)
```

```
for i in range(5):
```

```
    print(fib(i),end=" ")
```


lambda functions -

- * a lambda function is a small, anonymous (nameless) function in Python.
- * Defined using the keyword lambda instead of def.
- * Can take any number of arguments but must contain only one expression.
- * expression is automatically returned (no need to use return).

Syntax -

lambda arguments : expression.

lambda 5: 5*5 # we can't ~~hav~~ give directly we have to take the variable & use it

SyntaxError - invalid syntax.

Square = lambda x: x*x

Square(5)

25

add = lambda a, b: a+b

add(3, 6)

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Uses of lambda -

- Useful for short, simple operation
- Saves code (one-liners)
- often used with map(), filter(), or as inline function